

Exam Cheating Detection Report

27 Potential Cheating Incident(s) Detected

Video Duration	61m 24s
Original Resolution	1920x1080
Frame Rate	25.0 fps
Analysis Date	2026-02-04 14:57

Executive Summary

Analysis Overview:

- Total students identified: 22
- Teacher identified: Yes
- Potential cheating pairs detected: 27

Alert Summary

Students	Time Periods	Confidence	Tier	Primary Behaviors
R2_C5 & R2_C7	8:48-8:48, 12:22-12:32, 8:36-8:46...	84.8%	HIGH	Body leaning toward peer (24700 insta...
R2_C12 & R2_C12	16:46, 16:34-16:48, 16:30-16:43	78.7%	HIGH	Body leaning toward peer (18750 insta...
R2_C6 & R3_C1	1:34-1:44, 1:36-1:46, 1:31-1:42...	74.0%	HIGH	Body leaning toward peer (800 instanc...
R3_C2 & R2_C6	1:36-1:59, 1:34-1:46, 1:32-1:44...	73.5%	HIGH	Body leaning toward peer (550 instanc...
R2_C11 & R2_C11	11:50:45, 11:56-12:12, 11:54-12:09...	69.0%	MEDIUM	Body leaning toward peer (12200 insta...
R2_C12 & R2_C12	2:34-7:50, 2:52-3:04, 2:50-3:01...	68.3%	MEDIUM	Body leaning toward peer (18500 insta...
R2_C7 & R2_C8	1:30-1:46, 1:37-1:48, 1:28-1:43...	67.6%	MEDIUM	Body leaning toward peer (3250 instan...
R2_C6 & R2_C8	1:21-1:34, 1:25-1:36, 1:27-1:38...	67.1%	MEDIUM	Body leaning toward peer (800 instanc...
R2_C4 & R3_C1	0:36-1:00, 0:33-0:54, 0:26-0:47...	64.6%	MEDIUM	Body leaning toward peer (650 instanc...
R2_C4 & R2_C5	0:03-0:40, 0:32-0:44, 0:25-0:42...	63.4%	MEDIUM	Body leaning toward peer (600 instanc...
R2_C10 & R2_C7	11:15:04, 11:47-11:57, 11:49-11:56...	63.2%	MEDIUM	Body leaning toward peer (17000 insta...
R2_C4 & R2_C6	0:11-0:43, 0:19-0:46	63.0%	MEDIUM	Body leaning toward peer (100 instanc...
R2_C5 & R2_C8	1:35-1:47, 1:33-1:44, 1:38-1:50...	62.0%	MEDIUM	Body leaning toward peer (2950 instan...
R2_C14 & R2_C13	1:03-0:34, 1:14-1:28, 1:16-1:30...	61.8%	MEDIUM	Body leaning toward peer (2300 instan...
R2_C2 & R3_C3	1:14-3:25, 7:43-7:54, 3:20-3:38...	61.2%	MEDIUM	Body leaning toward peer (5750 instan...
R3_C4 & R2_C14	0:04-0:19, 0:21-0:33, 0:23-0:35...	61.2%	MEDIUM	Body leaning toward peer (2250 instan...
R2_C7 & R2_C6	0:22-0:45, 0:19-0:42, 0:11-0:38...	59.9%	MEDIUM	Body leaning toward peer (800 instanc...
R2_C5 & R2_C6	0:19-0:45, 0:32-1:22, 0:23-1:07...	59.8%	MEDIUM	Body leaning toward peer (750 instanc...
R2_C12 & R3_C3	0:40-1:01, 0:36-0:59, 0:31-0:50...	59.3%	MEDIUM	Body leaning toward peer (1250 instan...
R2_C12 & R4_C2	0:11-0:33, 0:05-0:26, 0:03-0:22	57.3%	MEDIUM	Body leaning toward peer (150 instanc...

R2_C11 & R2_C10	04-1:15, 1:01-1:13, 0:47-0:56...	56... 57.0%	MEDIUM	Body leaning toward peer (6600 instan...
R2_C9 & R2_C8	17-0:38, 0:10-0:24, 0:20-0:42...	42... 56.1%	MEDIUM	Body leaning toward peer (200 instanc...
R2_C4 & R3_C2	45-1:02, 0:11-0:52, 0:30-1:00...	00... 54.8%	LOW	Body leaning toward peer (300 instanc...
R2_C11 & R2_C10	13-1:17, 0:47-1:00, 1:10-1:19...	19... 52.2%	LOW	Body leaning toward peer (2050 instan...
R2_C10 & R2_C13	35-2:45, 2:31-2:41, 0:45-0:58...	58... 51.4%	LOW	Body leaning toward peer (2850 instan...
R2_C11 & R3_C1	12-1:26, 1:09-1:24, 0:42-1:02...	02... 50.6%	LOW	Body leaning toward peer (800 instanc...
R2_C11 & R2_C8	11-2:22, 2:13-2:25, 2:15-2:27...	27... 50.0%	LOW	Body leaning toward peer (3600 instan...

Detailed Alert Analysis

Alert 1: R2_C5 & R2_C7

Confidence Score: 84.8% (HIGH)

Suspicious Time Periods: 8:38 - 8:48, 12:22 - 12:32, 8:36 - 8:46, 8:40 - 8:52, 12:24 - 12:34

Detected Behaviors:

- Body leaning toward peer (24700 instances)
 - Hand activity near face (24695 instances)
 - Mutual gaze (looking at each other) (9255 instances)
 - Sustained head turn toward peer (6747 instances)
 - Whispering posture detected (6727 instances)
 - Moved closer than normal (210 instances)
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Alert 2: R2_C12 & R2_C10

Confidence Score: 78.7% (HIGH)

Suspicious Time Periods: 16:32 - 16:46, 16:34 - 16:48, 16:30 - 16:43, 16:42 - 16:52, 16:25 - 16:39

Detected Behaviors:

- Body leaning toward peer (18750 instances)
 - Hand activity near face (18750 instances)
 - Sustained head turn toward peer (14721 instances)
 - Whispering posture detected (14636 instances)
 - Moved closer than normal (7930 instances)
 - Mutual gaze (looking at each other) (4264 instances)
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Alert 3: R2_C6 & R3_C1

Confidence Score: 74.0% (HIGH)

Suspicious Time Periods: 1:34 - 1:44, 1:36 - 1:46, 1:31 - 1:42, 1:29 - 1:40, 1:38 - 1:58

Detected Behaviors:

- Body leaning toward peer (800 instances)
 - Moved closer than normal (800 instances)
 - Hand activity near face (683 instances)
 - Mutual gaze (looking at each other) (474 instances)
 - Sustained head turn toward peer (301 instances)
 - Whispering posture detected (194 instances)
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Alert 4: R3_C2 & R2_C6

Confidence Score: 73.5% (HIGH)

Suspicious Time Periods: 1:36 - 1:59, 1:34 - 1:46, 1:32 - 1:44, 1:30 - 1:41, 0:11 - 1:25

Detected Behaviors:

- Body leaning toward peer (550 instances)
 - Hand activity near face (408 instances)
 - Sustained head turn toward peer (384 instances)
 - Whispering posture detected (273 instances)
 - Mutual gaze (looking at each other) (226 instances)
 - Moved closer than normal (144 instances)
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Alert 5: R2_C11 & R2_C10

Confidence Score: 69.0% (MEDIUM)

Suspicious Time Periods: 0:35 - 0:45, 11:56 - 12:12, 11:54 - 12:09, 15:36 - 15:50, 11:58 - 12:14

Detected Behaviors:

- Body leaning toward peer (12200 instances)
 - Hand activity near face (12200 instances)
 - Sustained head turn toward peer (10150 instances)
 - Whispering posture detected (10100 instances)
 - Moved closer than normal (7625 instances)
 - Mutual gaze (looking at each other) (1836 instances)
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Alert 6: R2_C12 & R2_C11

Confidence Score: 68.3% (MEDIUM)

Suspicious Time Periods: 7:34 - 7:50, 2:52 - 3:04, 2:50 - 3:01, 2:54 - 3:07, 2:58 - 3:09

Detected Behaviors:

- Body leaning toward peer (18500 instances)
 - Hand activity near face (18490 instances)
 - Sustained head turn toward peer (16870 instances)
 - Whispering posture detected (16435 instances)
 - Moved closer than normal (5308 instances)
 - Mutual gaze (looking at each other) (1740 instances)
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Alert 7: R2_C7 & R2_C8

Confidence Score: 67.6% (MEDIUM)

Suspicious Time Periods: 1:30 - 1:46, 1:37 - 1:48, 1:28 - 1:43, 1:39 - 1:50, 1:26 - 1:41

Detected Behaviors:

- Body leaning toward peer (3250 instances)
 - Hand activity near face (3250 instances)
 - Sustained head turn toward peer (1064 instances)
 - Whispering posture detected (1064 instances)
 - Mutual gaze (looking at each other) (940 instances)
 - Moved closer than normal (1 instances)
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Alert 8: R2_C6 & R2_C8

Confidence Score: 67.1% (MEDIUM)

Suspicious Time Periods: 1:21 - 1:34, 1:25 - 1:36, 1:27 - 1:38, 0:46 - 1:31, 0:44 - 1:29

Detected Behaviors:

- Body leaning toward peer (800 instances)
 - Hand activity near face (800 instances)
 - Sustained head turn toward peer (557 instances)
 - Whispering posture detected (414 instances)
 - Mutual gaze (looking at each other) (374 instances)
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Alert 9: R2_C4 & R3_C1

Confidence Score: 64.6% (MEDIUM)

Suspicious Time Periods: 0:36 - 1:00, 0:33 - 0:54, 0:26 - 0:47, 0:21 - 0:40, 0:38 - 1:03

Detected Behaviors:

- Body leaning toward peer (650 instances)
 - Moved closer than normal (627 instances)
 - Sustained head turn toward peer (579 instances)
 - Hand activity near face (546 instances)
 - Whispering posture detected (281 instances)
 - Mutual gaze (looking at each other) (179 instances)
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Alert 10: R2_C4 & R2_C5

Confidence Score: 63.4% (MEDIUM)

Suspicious Time Periods: 0:03 - 0:40, 0:32 - 0:44, 0:25 - 0:42, 0:35 - 0:46, 0:40 - 0:51

Detected Behaviors:

- Body leaning toward peer (600 instances)
 - Hand activity near face (600 instances)
 - Sustained head turn toward peer (503 instances)
 - Whispering posture detected (464 instances)
 - Moved closer than normal (389 instances)
 - Mutual gaze (looking at each other) (80 instances)
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Alert 11: R2_C10 & R2_C7

Confidence Score: 63.2% (MEDIUM)

Suspicious Time Periods: 14:54 - 15:04, 11:47 - 11:57, 11:49 - 11:59, 19:52 - 20:06, 14:56 - 15:06

Detected Behaviors:

- Body leaning toward peer (17000 instances)
 - Hand activity near face (17000 instances)
 - Mutual gaze (looking at each other) (9138 instances)
 - Moved closer than normal (3046 instances)
 - Sustained head turn toward peer (2226 instances)
 - Whispering posture detected (2216 instances)
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Alert 12: R2_C4 & R2_C6

Confidence Score: 63.0% (MEDIUM)

Suspicious Time Periods: 0:11 - 0:43, 0:19 - 0:46

Detected Behaviors:

- Body leaning toward peer (100 instances)
 - Hand activity near face (97 instances)
 - Moved closer than normal (85 instances)
 - Sustained head turn toward peer (84 instances)
 - Whispering posture detected (57 instances)
 - Mutual gaze (looking at each other) (26 instances)
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Alert 13: R2_C5 & R2_C8

Confidence Score: 62.0% (MEDIUM)

Suspicious Time Periods: 1:35 - 1:47, 1:33 - 1:44, 1:38 - 1:50, 1:58 - 2:53, 1:31 - 1:42

Detected Behaviors:

- Body leaning toward peer (2950 instances)
- Hand activity near face (2950 instances)
- Sustained head turn toward peer (937 instances)

- Whispering posture detected (937 instances)
 - Mutual gaze (looking at each other) (581 instances)
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Alert 14: R2_C14 & R2_C13

Confidence Score: 61.8% (MEDIUM)

Suspicious Time Periods: 0:03 - 0:34, 1:14 - 1:28, 1:16 - 1:30, 1:19 - 1:32, 0:10 - 0:36

Detected Behaviors:

- Body leaning toward peer (2300 instances)
 - Hand activity near face (2300 instances)
 - Sustained head turn toward peer (2064 instances)
 - Whispering posture detected (2064 instances)
 - Mutual gaze (looking at each other) (130 instances)
 - Moved closer than normal (86 instances)
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Alert 15: R2_C2 & R3_C1

Confidence Score: 61.2% (MEDIUM)

Suspicious Time Periods: 3:14 - 3:25, 7:43 - 7:54, 3:20 - 3:38, 3:12 - 3:22, 3:16 - 3:27

Detected Behaviors:

- Body leaning toward peer (5750 instances)
 - Hand activity near face (5740 instances)
 - Sustained head turn toward peer (4597 instances)
 - Whispering posture detected (2906 instances)
 - Moved closer than normal (1332 instances)
 - Mutual gaze (looking at each other) (1252 instances)
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Alert 16: R3_C4 & R2_C14

Confidence Score: 61.2% (MEDIUM)

Suspicious Time Periods: 0:04 - 0:19, 0:21 - 0:33, 0:23 - 0:35, 0:25 - 0:37, 0:27 - 0:39

Detected Behaviors:

- Body leaning toward peer (2250 instances)
- Hand activity near face (2170 instances)
- Sustained head turn toward peer (2105 instances)
- Whispering posture detected (1908 instances)
- Mutual gaze (looking at each other) (82 instances)

Alert 17: R2_C7 & R2_C6

Confidence Score: 59.9% (MEDIUM)

Suspicious Time Periods: 0:22 - 0:45, 0:19 - 0:42, 0:11 - 0:38, 0:30 - 0:52, 0:32 - 1:22

Detected Behaviors:

- Body leaning toward peer (800 instances)
 - Hand activity near face (794 instances)
 - Sustained head turn toward peer (417 instances)
 - Mutual gaze (looking at each other) (392 instances)
 - Whispering posture detected (270 instances)
 - Moved closer than normal (144 instances)
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Alert 18: R2_C5 & R2_C6

Confidence Score: 59.8% (MEDIUM)

Suspicious Time Periods: 0:19 - 0:45, 0:32 - 1:22, 0:23 - 1:07, 0:39 - 1:24, 1:08 - 1:30

Detected Behaviors:

- Body leaning toward peer (750 instances)
 - Hand activity near face (750 instances)
 - Mutual gaze (looking at each other) (409 instances)
 - Sustained head turn toward peer (331 instances)
 - Whispering posture detected (213 instances)
 - Moved closer than normal (182 instances)
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Alert 19: R2_C12 & R3_C3

Confidence Score: 59.3% (MEDIUM)

Suspicious Time Periods: 0:40 - 1:01, 0:36 - 0:59, 0:31 - 0:50, 0:45 - 1:04, 1:04 - 1:19

Detected Behaviors:

- Body leaning toward peer (1250 instances)
 - Hand activity near face (1198 instances)
 - Sustained head turn toward peer (830 instances)
 - Whispering posture detected (717 instances)
 - Moved closer than normal (409 instances)
 - Mutual gaze (looking at each other) (82 instances)
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Alert 20: R2_C12 & R4_C2

Confidence Score: 57.3% (MEDIUM)

Suspicious Time Periods: 0:11 - 0:33, 0:05 - 0:26, 0:03 - 0:22

Detected Behaviors:

- Body leaning toward peer (150 instances)
 - Hand activity near face (141 instances)
 - Sustained head turn toward peer (135 instances)
 - Whispering posture detected (101 instances)
 - Moved closer than normal (63 instances)
 - Mutual gaze (looking at each other) (11 instances)
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Alert 21: R2_C11 & R2_C14

Confidence Score: 57.0% (MEDIUM)

Suspicious Time Periods: 1:04 - 1:15, 1:01 - 1:13, 0:47 - 0:56, 1:06 - 1:18, 0:45 - 0:54

Detected Behaviors:

- Body leaning toward peer (6600 instances)
 - Hand activity near face (6600 instances)
 - Mutual gaze (looking at each other) (1183 instances)
 - Sustained head turn toward peer (575 instances)
 - Whispering posture detected (575 instances)
 - Moved closer than normal (280 instances)
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Alert 22: R2_C9 & R2_C8

Confidence Score: 56.1% (MEDIUM)

Suspicious Time Periods: 0:17 - 0:38, 0:10 - 0:24, 0:20 - 0:42, 0:15 - 0:35

Detected Behaviors:

- Body leaning toward peer (200 instances)
 - Hand activity near face (195 instances)
 - Sustained head turn toward peer (149 instances)
 - Whispering posture detected (106 instances)
 - Mutual gaze (looking at each other) (63 instances)
 - Moved closer than normal (3 instances)
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Alert 23: R2_C4 & R3_C2

Confidence Score: 54.8% (LOW)

Suspicious Time Periods: 0:45 - 1:02, 0:11 - 0:52, 0:30 - 1:00, 0:47 - 1:05, 0:17 - 0:55

Detected Behaviors:

- Body leaning toward peer (300 instances)
 - Sustained head turn toward peer (291 instances)
 - Hand activity near face (254 instances)
 - Moved closer than normal (190 instances)
 - Whispering posture detected (88 instances)
 - Mutual gaze (looking at each other) (39 instances)
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Alert 24: R2_C11 & R2_C13

Confidence Score: 52.2% (LOW)

Suspicious Time Periods: 1:08 - 1:17, 0:47 - 1:00, 1:10 - 1:19, 1:12 - 1:21, 0:52 - 1:05

Detected Behaviors:

- Body leaning toward peer (2050 instances)
 - Hand activity near face (2050 instances)
 - Mutual gaze (looking at each other) (713 instances)
 - Sustained head turn toward peer (391 instances)
 - Whispering posture detected (391 instances)
 - Moved closer than normal (3 instances)
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Alert 25: R2_C10 & R2_C13

Confidence Score: 51.4% (LOW)

Suspicious Time Periods: 2:35 - 2:45, 2:31 - 2:41, 0:45 - 0:58, 0:48 - 1:01, 0:50 - 1:03

Detected Behaviors:

- Body leaning toward peer (2850 instances)
 - Hand activity near face (2850 instances)
 - Mutual gaze (looking at each other) (1447 instances)
 - Moved closer than normal (745 instances)
 - Sustained head turn toward peer (458 instances)
 - Whispering posture detected (458 instances)
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Alert 26: R2_C11 & R3_C3

Confidence Score: 50.6% (LOW)

Suspicious Time Periods: 1:12 - 1:26, 1:09 - 1:24, 0:42 - 1:02, 0:37 - 0:59, 1:07 - 1:22

Detected Behaviors:

- Body leaning toward peer (800 instances)
 - Hand activity near face (785 instances)
 - Mutual gaze (looking at each other) (597 instances)
 - Sustained head turn toward peer (114 instances)
 - Whispering posture detected (41 instances)
 - Moved closer than normal (16 instances)
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Alert 27: R2_C11 & R2_C8

Confidence Score: 50.0% (LOW)

Suspicious Time Periods: 2:11 - 2:22, 2:13 - 2:25, 2:15 - 2:27, 2:17 - 2:29, 2:21 - 2:31

Detected Behaviors:

- Body leaning toward peer (3600 instances)
- Hand activity near face (3600 instances)
- Mutual gaze (looking at each other) (2187 instances)
- Sustained head turn toward peer (8 instances)
- Whispering posture detected (8 instances)

Methodology

This analysis was conducted using a 7-stage computer vision pipeline:

1. Video Preprocessing

Frames extracted at 5 fps with CLAHE contrast enhancement in LAB color space.

2. Person Detection & Tracking

YOLOv8 object detection with BoT-SORT multi-object tracking for consistent student identification throughout the video.

3. Pose Estimation

YOLOv8-Pose model extracts 17 COCO body keypoints. Head orientation is computed using solvePnP with facial landmarks.

4. Spatial Analysis

Automatic seat assignment based on median positions during warm-up period. Neighbor graphs built using distance thresholds. Teachers identified by movement patterns and standing posture.

5. Behavioral Feature Extraction

Per-frame features include: head direction, mutual gaze detection, whispering posture, body lean, hand position relative to face and neighbors, paper sharing gestures.

6. Temporal Classification

Sliding window analysis (10-second windows, 2-second stride) with weighted scoring. Multi-criteria filter requires: minimum suspicious windows, confidence threshold, mutual behavior evidence, and absence of nearby teacher.

7. Alert Generation

Confidence tiers (Very High > 85%, High > 70%, Medium > 55%) with timestamped evidence and behavior summaries.

Important Notes:

- This system is designed to flag potential incidents for human review.
- False positives may occur due to normal classroom interactions.
- Results should be verified by reviewing the evidence frames and video.
- Teacher proximity during flagged periods may indicate supervised activity.